

UCT632: Modern Web Applications

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3	0	2	4.0

Course Objectives: To provide an understanding of various modern programming languages such as HTML, CSS, JavaScript, XML, PHP and MySQL for the Web development.

Introduction to Internet & World Wide Web: History of the Internet & World- Wide Web, Web Browsers, Web Servers, Uniform Resource Locator, Tools and Web Programming Languages, Web Standards, Categories of Web Applications, Characteristics of Web Applications, Tiered Architecture.

Hypertext Mark Up Language: Basic HTML page, Text Formatting, Table, Headers, Linking, Images, List, Meta Elements.

Cascading Style Sheets (CSS): Inline, Internal and External Style Sheet, Bootstrap-CSS Text, CSS forms, CSS components drop down.

Java Script: Introduction to Java Scripts, Objects in Java Script, Dynamic HTML with Java Script, Bootstrap- JS Alert, JS Button, JS popover.

Extensible Markup Language (XML): Introduction, Structuring Data, Document Type Definition, XML Vocabularies, Document Object Model (DOM) with JavaScript, Extensible Stylesheet Language Transforms (XSL).

Writing Basic PHP Programs: Creating PHP Programs, Numbers and Strings, Literals Variables, Operators and Functions.

Form & PHP: Creating Form Controls, Using Values Returned From, Forms Using PHP

PHP Database Connectivity: Connecting to MySQL Server, Selecting Databases, Checking for Errors, Closing the MySQL Server Connection Manipulating Data in MySQL Using PHP: Inserting, Viewing, Updating and Deleting Records, Manipulating joined tables.

User Authentication: Creating Session, Authorization Level

Laboratory Work:

Use of tags in HTML ,Creating and designing Form , Applying Style rules with CSS, Positioning and Layout in CSS, JavaScript Variables, Operators, Array, Strings and Functions ,Objects in Java Script, XML and DOM, Event Handling with PHP, Creating and Manipulating Data in MySQL.

Course Learning Outcomes (CLOs) / Course Objectives (COs):

Upon completion of this course, students will be able to:

1. Comprehend the concept of the Web Server, Web Browser, URL and usage of Web Programming Languages for the Web development.
2. Illustrate the usage of tags and style rules in designing or styling of the Web Pages through HTML and CSS.
3. Analyze the role of XML and JavaScript in creating the interactive web pages on the Internet.
4. Understand the need of Server Side Scripting Language PHP and MYSQL Databases in the development of the Web Applications.

Text Books:

1. Deitel P. J., Deitel H. M. and Deitel A. (2012) Internet and World Wide Web: How to Program, 5th Edition, Pearson Prentice Hall.
2. HTML & CSS: Design and Build Websites, Jon Duckett, John Wiley & Sons.
3. Naramore E., Gerner J., Scouarnec Y.L., et al., (2005) Beginning PHP5, Apache, MySQL Web Development: Programmer to Programmer, John Wiley & Sons Inc.

Reference Books:

1. Sebesta R. W. (2014) Programming the World Wide Web, 8th edition, Pearson.
2. Pressman R. and Lowe D. (2008) Web Engineering: a practitioner's approach, 1st Edition, McGraw Hill.
3. Kappel G., et al. (2006) Web Engineering: The Discipline of systematic Development of Web Applications, 1st Edition, John Wiley & Sons.
4. Suh W. (2005) Web Engineering: Principles and Techniques, Idea Group Inc.
5. Ullman L (2016) PHP for the Web: Visual Quick Start Guide, 5th Edition, Peachpit Press.